

Video Presentation Script

Target length: about 2 minutes 40 seconds

Contactless Radar-Based Heart Rate Monitoring Device

Text / Voiceover	Action on Screen	Time
none	Static title slide with minimal text: Contactless Radar-Based Heart Rate Monitoring Device , ZHAW PM4 Electronics Project	0:00-0:06
Heart rate tells a lot. But measuring it often still means contact: electrodes, straps, watches, or wearable sensors.	Problem visuals: ECG electrodes, smartwatch, chest strap. Minimal text: Heart rate. Still contact-based.	0:07-0:18
For sleep, long-term monitoring, or sensitive users, contact can become the limitation. The ideal measurement is comfortable, continuous, and almost invisible.	Sleeping person, patient monitoring, contact-free symbol. Minimal text: comfortable , continuous , contact-free	0:19-0:32
This device detects heart-rate-related chest movement without touching the body. It processes the radar signal in real time on an STM32 microcontroller and shows the result directly on an integrated display.	Product reveal / feature slide: hardware photo or clean render. Short labels: Contactless , Real-time , LCD Touchscreen Display , ECG reference	0:33-0:48
The principle is Doppler sensing. A microwave radar module sends a signal toward the chest. Small movements caused by cardiac activity change the phase of the reflected signal, creating a measurable heartbeat pattern.	Animation: radar beam, reflected signal, moving chest, signal wave.	0:49-1:05
Inside the system, the K-LC5 radar module is connected to an analog front end and an STM32 Discovery board. The microcontroller acquires the I and Q radar channels, processes them, and drives the LCD touchscreen.	Block diagram: K-LC5 radar -> analog front end -> STM32 -> LCD touchscreen . Brief real-hardware shot.	1:06-1:22
The complete processing chain runs directly on the embedded platform. No external PC is required during operation, while heart-rate data can also be logged to an SD card with the integrated OpenLog module for later analysis.	Live video of standalone board operation. Brief close-up of OpenLog. Optional text: Standalone operation , SD card logging	1:23-1:39

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The radar signals are sampled at 100 hertz. Each analysis window contains 512 samples per channel, equal to 5.12 seconds of data, with updates every 2.56 seconds using 50 percent overlap.	Processing pipeline slide with key numbers: 100 Hz , 512 samples , 5.12 s window , 2.56 s update	1:40-1:55
From the sampled I and Q signals, the system computes the frequency spectrum and detects the dominant heart-rate peak. The interface also provides analysis tools such as selectable filter modes for testing and development.	UI recording: time-domain screen, FFT spectrum, peak detection. Briefly show filter/effect menu only for 1–2 seconds.	1:56-2:12
The interface makes the signal processing visible: time signal, FFT spectrum, peak-frequency readout, ECG monitor page for reference comparison, and logged heart-rate data for offline evaluation.	Live UI recording. Show: Time Signal , FFT Spectr , Peak Detect , EKG BPM . Optional quick insert: SD card/log file on PC.	2:13-2:30
This allows the user to see both the final heart-rate estimate and the intermediate signal and spectrum, supporting debugging, validation, and further development.	Continue UI video. Zoom/crop on spectrum peak and ECG BPM page.	2:31-2:43
The device targets short-range operation up to about one meter. The displayed spectrum covers approximately plus or minus four hertz, with ECG comparison used for validation.	Static technical summary slide: Range: up to ~1 m , Spectrum: +/-4 Hz , ECG comparison , SD logging	2:44-2:55
The FFT bin spacing is approximately 0.195 hertz, or 11.7 beats per minute. However, parabolic peak interpolation refines the peak position, so the BPM output is not limited to raw FFT-bin steps. Practical accuracy still depends on signal quality, leakage, motion artifacts, and temporal stability.	Technical graphic: FFT peak between bins, parabolic curve over three bins. Labels: 0.195 Hz , 11.7 bpm/bin , sub-bin interpolation , accuracy depends on signal quality	2:56-3:16
Contactless sensing, real-time embedded processing, touchscreen visualization, ECG-supported validation, and SD-card logging come together in one compact platform for comfortable vital-sign monitoring.	**Final slide:** project title, authors, date, ZHAW affiliation, repository/project link or QR code.	3:17-3:27

Final Slide Content

Use the last slide to repeat the required summary in compact form:

- Contactless heart-rate monitoring
- Short-range operation: up to about 1 m
- Real-time embedded processing on STM32F429
- Touchscreen UI with time, FFT, peak, and ECG pages
- FFT bin spacing: about 0.195 Hz
- Accuracy: add measured ECG comparison result here
- Authors: Thomas Perri, Bogdans Grebnevs
- Date: add recording or submission date
- Link: add repository or demo link